

Dissertation Proposal

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Outline

1. Changes-in-Changes with a Time Varying Treatment
2. Differences-in-Differences in the Presence of Mediators
3. Anticipation Effects in Differences-in-Differences

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1. Changes-in-Changes with a Time Varying Treatment
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3. Anticipation Effects in Differences-in-Differences

Motivation

- ▶ Changes-in-Changes (CiC) is a nonlinear extension of Difference-in-Differences (DiD) used to evaluate the effects of policy interventions or treatments (Athey and Imbens, 2006)
- ▶ CiC is designed for binary treatment
- ▶ Many empirical settings involve continuous treatment
- ▶ Example: Marginal propensity to consume
Changes in disposable income $\rightarrow$ Non-durable consumption
- ▶ Question: Can we extend CiC to continuous treatment?

This Paper

Setting:



Results :

- ▶ By restricting the dependence structure of unobservables,
we identify the counterfactual outcome distribution under different treatment levels

Related Literature

- ▶ DiD with continuous treatment

Callaway, Goodman-Bacon, and Sant'Anna (2024, 2025); de Chaisemartin, D'Haultfœuille, Pasquier, Sow, and Vazquez-Bare (2022); de Chaisemartin, D'Haultfœuille, and Vazquez-Bare (2024)

This paper: Treatment can be nonzero in both periods; No naturally defined control group/stayers

- ▶ IV with discrete support

D'Haultfœuille and Février (2015); Torgovitsky (2015); Chernizhukov, Fernández-Val, and Luo (2025); Chernizhukov, Fernández-Val, Han, and Wüthrich (2024)

This paper: Need time to provide exogenous variation in treatment, and allow it to directly affect outcome. (Allow the exclusion restriction to be violated)

Setup and Notation: Panel Data

- ▶ Unit are drawn from an infinite superpopulation and observed over two periods $t \in \{0, 1\}$
- ▶ Each unit can be an individual, firm, or geographical unit (county or region).
- ▶ Let $D_{it} \in \mathcal{D}_t$ denote a continuous treatments at period t , and $\mathbf{D}_i = (D_{i0}, D_{i1}) \in \mathcal{D}_{01}$
Let $S_i = \mathbb{1}\{D_{i0} \neq D_{i1}\}$ be an indicator equal to 1 if the unit's treatment changes over two periods
- ▶ Let $Y_{it}(\mathbf{d}_i)$ denote potential outcomes at period t under treatment path $\mathbf{d}_i = (d_{i0}, d_{i1}) \in \mathcal{D}_{01}$
Static effect: $Y_{it}(\mathbf{d}_i) = Y_{it}(d_{it})$, $\forall \mathbf{d}_i = (d_{i0}, d_{i1})$ and $t \in \{0, 1\}$
Consistency: the observed outcome satisfies $Y_{it} = Y_{it}(D_{it})$
- ▶ Suppress the individual index i for simplicity

Parameter of Interest

Interested in $F_{Y_t(d)}$ for $d \in \mathcal{D}_t$ and its functionals , e.g.,

- ▶ Average structural functions $ASF_t(d) := \mathbb{E}[Y_t(d)]$

Average treatment effects $ATE_t(d, d') := ASF_t(d) - ASF_t(d')$ for all $d, d' \in \mathcal{D}_t$

- ▶ Quantile structural functions: $QSF_t(d, \tau) := Q_{Y_t(d)}(\tau)$

Quantile treatment effects: $QTE(\tau, d, d') := QSF_t(d, \tau) - QSF_t(d', \tau)$ for all $d, d' \in \mathcal{D}_t$

Framework

Assumption 0 (i.i.d. Sample)

We observe $\{(Y_{i0}, D_{i0}, Y_{i1}, D_{i1})\}_{i=1}^n$ that are independent and identically distributed vectors with the same probability distribution as (Y_0, D_0, Y_1, D_1)

Assumption 1 (Rank Invariance on the Time Trend)

$$Y_t(d) = g_t(\tilde{Y}_t(d))$$

and $\tilde{Y}_t(d) \in \mathbb{R}$, and $y \mapsto g_t(y)$ is strictly increasing.

- ▶ $\tilde{Y}_t(d)$: detrended outcome

$\tilde{Y}_t(d) = h(d, U_t)$ where U_t is multidimensional unobserved heterogeneity, e.g., preference, ability

- ▶ $g_t(y)$: Time effects/shocks, e.g., macro shocks

Time effect is nonlinear, and it preserves the ranking of individuals in outcome distribution

- ▶ Athey and Imbens (2006): untreated potential outcome $Y_t(0) = g_t(U_t)$

Framework

Assumption 2 (Time Homogeneity)

For all d and $\mathbf{d}$,

$$\tilde{Y}_0(d) \mid \mathbf{D} = \mathbf{d} \sim \tilde{Y}_1(d) \mid \mathbf{D} = \mathbf{d}$$

- ▶ Individuals with the same treatment path have the same distribution of $\tilde{Y}_t(d)$
- ▶ Suppose $\tilde{Y}_t(d) = h(d, U_t)$, A2 is equivalent to $U_0 \mid \mathbf{D} = \mathbf{d} \sim U_1 \mid \mathbf{D} = \mathbf{d}$
- ▶ Athey and Imbens (2006): $U_0 \mid D = d \sim U_1 \mid D = d$ for $d \in \{0, 1\}$

Assumption 3 (Support Condition)

There exists a positive measure of stayers, i.e., $0 < \mathbb{P}(S = 1) < 1$

Identification: Time Trend $g_t(\cdot)$

- ▶ Given the strict monotonicity, normalize $g_1(y) = y$ so that

$$Y_0(d) = g_0(\tilde{Y}_0(d)) \quad Y_1(d) = \tilde{Y}_1(d)$$

- ▶ By using stayers $D_0 = D_1$,

$$g_0(\cdot) = F_{Y_0|S}^{-1}\left(F_{Y_1|S}(\cdot \mid 0) \mid 0\right)$$

where $S = \mathbb{1}\{D_0 \neq D_1\}$

- ▶ Obtain the detrended outcome $\tilde{Y}_t = g_t^{-1}(Y_t)$ and $\tilde{Y}_t(d) = g_t^{-1}(Y_t(d))$

Local Gaussian Representation

Lemma (Chernozhukov et al., 2025)

Given $t \in \{0, 1\}$ and $d^* \in \mathcal{D}_t$, the joint distribution $F_{\tilde{Y}_t(d^*), D_t}$ can be represented by a standard bivariate normal distribution at any point (y, d) as

$$F_{\tilde{Y}_t(d^*), D_t}(y, d) = \Phi_2\left(\Phi^{-1}\left(F_{\tilde{Y}_t(d^*)}(y)\right), \Phi^{-1}\left(F_{D_t}(d)\right); \rho_{\tilde{Y}_t(d^*), D_t}(y, d)\right)$$

where $\Phi_2(\cdot, \cdot; \rho)$ is the standard bivariate normal CDF with parameter ρ , and $\rho_{\tilde{Y}_t(d^*), D_t}(y, d) \in [-1, 1]$ is the local correlation parameter, whose value is unique. 

Local Gaussian Representation

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where $\Phi_2(\cdot, \cdot; \rho)$ is the standard bivariate normal CDF with parameter ρ , and $\rho_{\tilde{Y}_t(d^*), D_t}(y, d) \in [-1, 1]$ is the local correlation parameter, whose value is unique. 

- ▶ $\Phi_2\left(\Phi^{-1}(\cdot), \Phi^{-1}(\cdot), \rho\right)$ captures the dependence structure of $\tilde{Y}_t(d^*)$ and D_t
 - It is called Gaussian copula if $\rho(y, d) = \rho$ for all (y, d)
- ▶ $\rho(y, d)$ measure the local dependence of $\tilde{Y}_t(d^*)$ and D_t at (y, d)

Local Gaussian Representation

Assumption 4 (Stability of Gaussian Copula)

Fix $d^* \in \mathcal{D}$, consider the dependence structure between $\tilde{Y}_t(d^*)$ and D_t ,

(i) The copula between $\tilde{Y}_t(d^*)$ and D_t is Gaussian.

i.e., $\rho_{\tilde{Y}_t(d^*), D_t}(y, d)$ does not depend on (y, d) , denoted by $\rho_{d^*, t}$

(ii) The copula between $\tilde{Y}_t(d^*)$ and D_t is stable over time, i.e., $\rho_{d^*, 0} = \rho_{d^*, 1} \equiv \rho_{d^*}$

- ▶ ρ_{d^*} : dependence parameter between $\tilde{Y}_t(d^*)$ and D_t
- ▶ The assumption restricts only the dependence structure, not marginals
- ▶ Testable when more than two periods are available

Identification: Panel Data

Assumption 4 implies

$$\begin{aligned}\Phi^{-1}\left(F_{\tilde{Y}_0|D_0}(y \mid d)\right) &= a_{d,y} + b_{d,y}\Phi^{-1}\left(F_{D_0}(d)\right) \\ \Phi^{-1}\left(F_{\tilde{Y}_1|D_1}(y \mid d)\right) &= a_{d,y} + b_{d,y}\Phi^{-1}\left(F_{D_1}(d)\right)\end{aligned}$$

where

$$\begin{aligned}a_{d,y} &= \frac{\Phi^{-1}\left(F_{\tilde{Y}_1(d)}(y)\right)}{\sqrt{1 - \rho_d(y)^2}} \\ b_{d,y} &= -\frac{\rho_d(y)}{\sqrt{1 - \rho_d(y)^2}}\end{aligned}$$

Proposition

Under Assumption 1 – 4, $F_{Y_1(d)}(y)$ and $\rho_d(y)$ are identified for $y \in \mathcal{Y}$ and $d \in \mathcal{D} \setminus d^*$.

Discussion

Example

Random coefficient model:

$$Y_{it}(d) = g_t\left(\alpha(\varepsilon_{it}) + \beta(\varepsilon_{it})d\right)$$

$$D_{it} = h_t(\eta_{it})$$

where $g_t(\cdot)$ and $h_t(\cdot)$ are strictly increasing, the joint distribution of $(\varepsilon_{it}, \eta_{it})$ is constant over time, and $\exists \eta^*$ s.t. $h_0(\eta^*) = h_1(\eta^*)$.

► Assumptions 1-3 hold with

$$U_t(d) = \alpha(\varepsilon_{it}) + \beta(\varepsilon_{it})d$$

$$d_{\text{cross}}^* = h_0(\eta^*)$$

Setup: Repeated Cross-Section Data

- Observe two repeated cross sections $\{(Y_{i0}, D_{i0})\}_{i=1}^n, \{(Y_{i1}, D_{i1})\}_{i=1}^n$

Existing Approaches

D'Haultfoeuille, Hoderlein, and Sasaki (2023)

- ▶ Extends Changes-in-Changes (Athey and Imbens, 2006) to continuous treatment
- ▶ Limitation: Only certain treatment effects on the treated are identified
E.g., for individuals with treatment d , the effects of moving from d to a specific d' can be identified
- ▶ Solutions proposed in DHS (2023)
 - (1) Shape restrictions + three periods $\rightarrow$ partial identification
 - (2) Linear functional form $\rightarrow$ identification of average treatment effects only

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Goal of this paper: To point identify richer treatment effects with only two periods.

Assumption 2 (Stationarity of Unobservables)

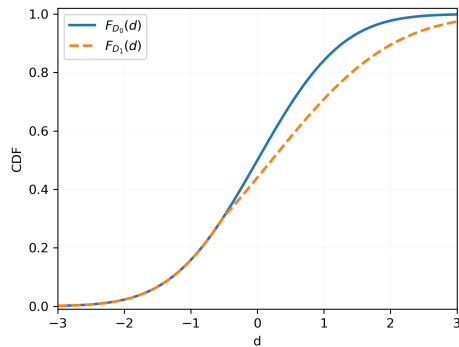
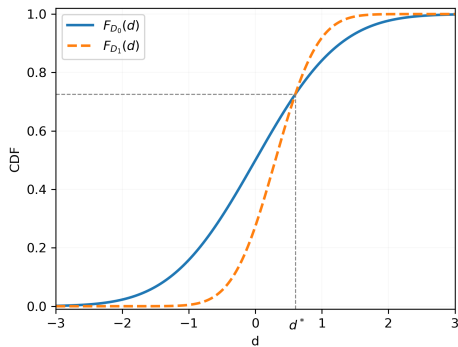
For all $(d, v) \in \mathcal{D} \times [0, 1]$,

$$U_0(d) \mid V_0 = v \sim U_1(d) \mid V_1 = v$$

- ▶ Individuals with the same treatment rank at two periods have the same distribution of $U_t(d)$
- ▶ Athey and Imbens (2006): $U_0 \mid D = d \sim U_1 \mid D = d$ for $d \in \{0, 1\}$

Assumption 3 (Crossing Points)

There exists a d^* s.t. $F_{D_0}(d^*) = F_{D_1}(d^*) \in (0, 1)$



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There exists a d^* s.t. $F_{D_0}(d^*) = F_{D_1}(d^*) \in (0, 1)$

- ▶ Exogenous change (e.g., a policy change) affects individuals' treatment in a heterogeneous way
- ▶ Observations with treatment level d^* serve as a control group
- ▶ Directly testable in the data

Example 1

Linear model:

$$Y_{it}(d) = \alpha_t + \theta d + \varepsilon_{it}$$

$$D_{it} = \gamma_t + \delta_t \eta_{it}$$

where $(\alpha_t, \gamma_t, \delta_t, \theta)$ are constants, the joint distribution of $(\varepsilon_{it}, \eta_{it})$ is constant over time.

► Assumptions 1-3 hold with

$$U_{it}(d) = \theta d + \varepsilon_{it}$$

$$g_t(u) = \alpha_t + u$$

$$d_{\text{cross}}^* = \frac{\delta_1 \gamma_0 - \delta_0 \gamma_1}{\delta_1 - \delta_0}$$

Example 2

Random coefficient model:

$$Y_{it}(d) = g_t(\alpha(\varepsilon_{it}) + \beta(\varepsilon_{it})d)$$

$$D_{it} = h_t(\eta_{it})$$

where $g_t(\cdot)$ and $h_t(\cdot)$ are strictly increasing, the joint distribution of $(\varepsilon_{it}, \eta_{it})$ is constant over time, and $\exists \eta^*$ s.t. $h_0(\eta^*) = h_1(\eta^*)$.

► Assumptions 1-3 hold with

$$U_t(d) = \alpha(\varepsilon_{it}) + \beta(\varepsilon_{it})d$$

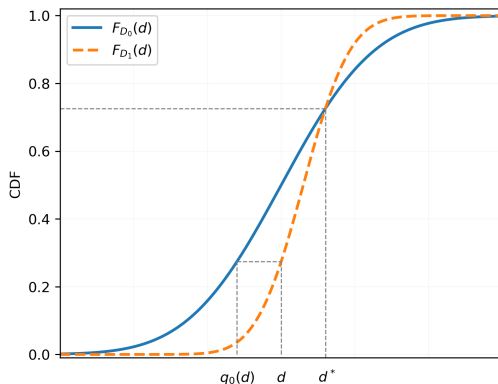
$$d_{\text{cross}}^* = h_0(\eta^*)$$

Framework

Under Assumptions 1–3, we can identify

$$ATT_1(d, q_0(d)) := \mathbb{E}[Y_1(q_0(d)) - Y_1(d) \mid D_1 = d]$$

where $q_0(d) = F_{D_0}^{-1}(F_{D_1}(d))$ is the treatment level in period 0 that corresponds to the same rank as d in period 1. $ATT_1(d, d')$ for arbitrary d' is not identified.



Identification

Under Assumption 1 and 2, for all $d \in \mathcal{D}$,

$$F_{\tilde{Y}_0(d), V_0}(y, v) = F_{\tilde{Y}_1(d), V_1}(y, v) \quad \forall (y, v) \in \mathcal{Y} \times [0, 1]$$

where $F_{\tilde{Y}_t(d), V_t}$ is the joint CDF of $\tilde{Y}_t(d)$ and $V_t = F_{D_t}(D_t)$.

- ▶ Joint distribution of $\tilde{Y}_t(d)$ and V_t does not change over time.
- ▶ Allow for arbitrary dependence between $\tilde{Y}_t(d)$ and V_t .

Example 1 (Continued)

Consider

$$Y_{it}(d) = \alpha_t + \theta d + \varepsilon_{it}$$

$$D_{it} = \gamma_t + \delta_t \eta_{it}$$

where $(\alpha_t, \gamma_t, \delta_t, \theta)$ are constants, the distribution of $(\varepsilon_{it}, \eta_{it})$ is constant over time.

- If $(\varepsilon_{it}, \eta_{it})$ are jointly normal with correlation ρ , then Assumption 4 holds.

Example 1 (Continued)

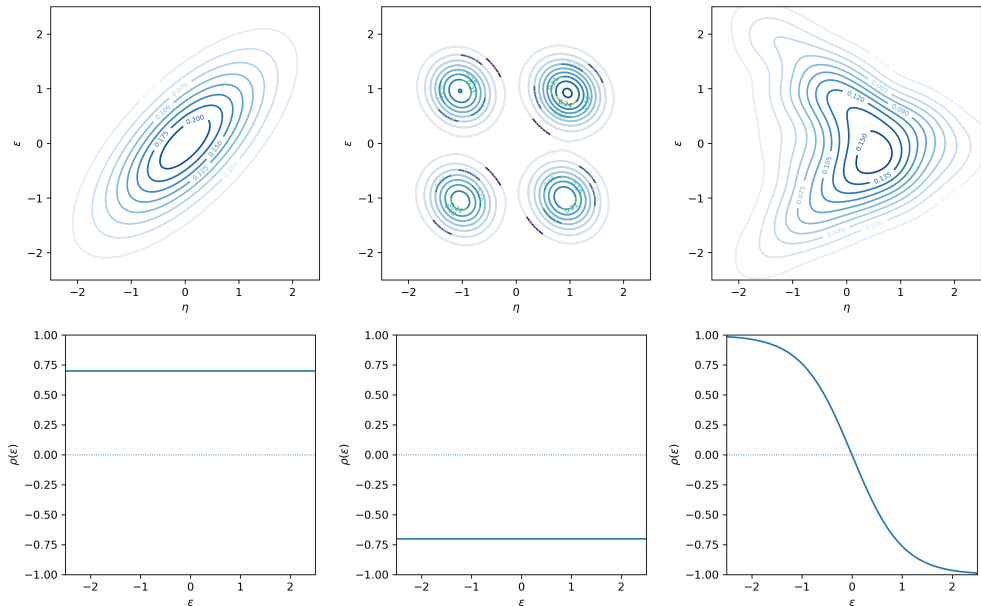


Figure: Contour plots of the joint density of ϵ_{it} and η_{it}

Identification

Given y and d ,

- In period 0, observe

$$F_{D_0}(d) = v_0$$

$$F_{\tilde{Y}_0|D_0}(y|d) = F_{\tilde{Y}_0(d)|v_0}(y|v_0)$$

- In period 1, observe

$$F_{D_1}(d) = v_1$$

$$F_{\tilde{Y}_1|D_1}(y|d) = F_{\tilde{Y}_1(d)|v_1}(y|v_1)$$

- To obtain $F_{\tilde{Y}_1(d)}(y)$, we need to extrapolate across rank $v \in (0, 1)$

Identification

Let $V_{it} = F_{D_{it}}(D_{it}) \in (0, 1)$ be the rank of treatment D_{it}

For $t \in \{0, 1\}$, consider

$$F_{\tilde{Y}_t|D_t}(y | d) = F_{\tilde{Y}_t(d)|D_t}(y | d) = F_{\tilde{Y}_t(d)|V_t}(y | F_{D_t}(d))$$

By the properties of the conditional distribution

$$\begin{aligned} F_{\tilde{Y}_t(d)|V_t}(y | v) &= \frac{\frac{\partial}{\partial v} F_{\tilde{Y}_t(d), V_t}(y, v)}{\frac{\partial}{\partial v} F_{V_t}(v)} \\ &= C_v^G \left(F_{\tilde{Y}_t(d)}(y), v, \rho_{\tilde{Y}_t(d), V_t}(y, v) \right) \\ &\quad + C_\rho^G \left(F_{\tilde{Y}_t(d)}(y), v, \rho_{\tilde{Y}_t(d), V_t}(y, v) \right) \frac{\partial}{\partial v} \rho_{\tilde{Y}_t(d), V_t}(y, v) \end{aligned}$$

To Do List

Local Gaussian Representation

The joint distribution $F_{X,Y}$ can be represented by a standard bivariate normal distribution at any point (x, y) as

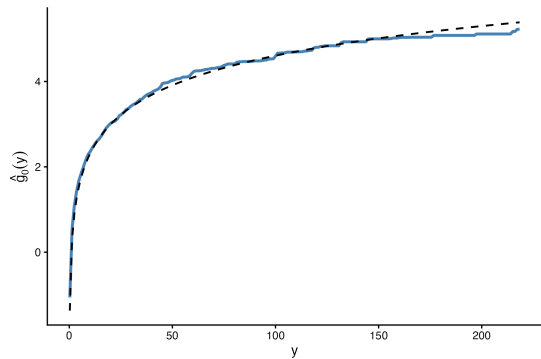
$$F_{X,Y}(x, y) = \Phi_2\left(\Phi^{-1}(F_X(x)), \Phi^{-1}(F_Y(y)) ; \rho_{X,Y}(x, y)\right)$$

where $\Phi_2(\cdot, \cdot; \rho)$ is the standard bivariate normal CDF with parameter ρ , and $\rho_{X,Y}(x, y) \in [-1, 1]$ is the unique solution in ρ to

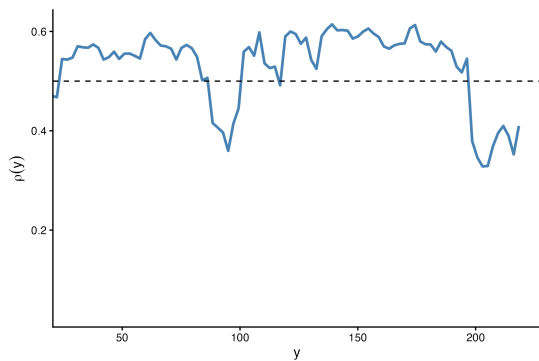
$$F_{X,Y}(x, y) = \Phi_2\left(\Phi^{-1}(F_X(x)), \Phi^{-1}(F_Y(y)) ; \rho\right)$$



Illustration



Time trend $g_0(y)$



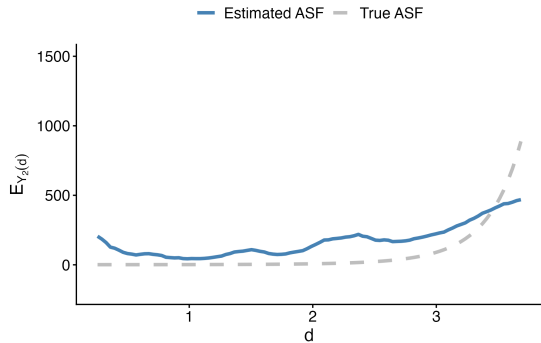
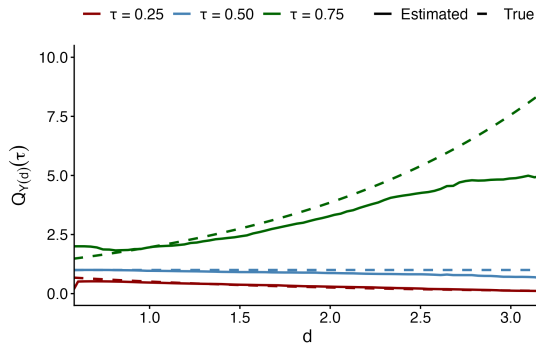
Local correlation



Illustration

$$Y_{i0}(d) = \varepsilon_{i0}d, \quad D_{i0} = 1 + \eta_{i0}$$

$$Y_{i1}(d) = \exp(\varepsilon_{i1}d), \quad D_{i1} = 2 + 2\eta_{i1}$$

Time trend $g_0(y)$ 

Quantile structural function

Single-index restriction

- Assumption 4 is equivalent to the single index restriction:

For each $d \in \mathcal{D}$,

$$F_{\tilde{Y}(d)|V}(y \mid v) = \Phi\left(a_{d,y} + b_{d,y}\Phi^{-1}(v)\right)$$

for $(y, v) \in \tilde{\mathcal{Y}} \times [0, 1]$, where

$$a_{d,y} = \frac{\Phi^{-1}\left(F_{\tilde{Y}(d)}(y)\right)}{\sqrt{1 - \rho_d(y)^2}}$$
$$b_{d,y} = -\frac{\rho_d(y)}{\sqrt{1 - \rho_d(y)^2}}$$

Estimation

- Step 1: Estimate the crossing point d^*

$$\hat{d}^* = \arg \min_{d \in [\hat{F}_{D_0}^{-1}(\underline{p}), \hat{F}_{D_1}^{-1}(\bar{p})]} \left| \hat{F}_{D_1}(d) - \hat{F}_{D_0}(d) \right|$$

- Step 2: Estimate the time trend g_0

$$\hat{g}_0(y) = \hat{F}_{Y_0|D_0}^{-1} \left[\hat{F}_{Y_1|D_1}(y | \hat{d}^*) | \hat{d}^* \right]$$

where

$$\hat{F}_{Y_t|D_t}(y | d) = \frac{\sum_{i=1}^n \mathbb{1}\{Y_{it} \leq y\} K\left(\frac{d - D_{it}}{h_n}\right)}{\sum_{i=1}^n K\left(\frac{d - D_{it}}{h_n}\right)}$$

and $K(\cdot)$ a kernel function and h_n is a bandwidth.

Detrend in period 0: $\tilde{Y}_0 = \hat{g}_0^{-1}(Y_0)$

Estimation

- Step 3: Solve the linear system

$$\hat{a}_{d,y} = \frac{\Phi^{-1}\left(\hat{F}_{\tilde{Y}_0|D_0}(y \mid d)\right) \Phi^{-1}\left(\hat{F}_{D_1}(d)\right) - \Phi^{-1}\left(\hat{F}_{Y_1|D_1}(y \mid d)\right) \Phi^{-1}\left(\hat{F}_{D_0}(d)\right)}{\Phi^{-1}\left(\hat{F}_{D_1}(d)\right) - \Phi^{-1}\left(\hat{F}_{D_0}(d)\right)}$$
$$\hat{b}_{d,y} = \frac{\Phi^{-1}\left(\hat{F}_{Y_1|D_1}(y \mid d)\right) - \Phi^{-1}\left(\hat{F}_{\tilde{Y}_0|D_0}(y \mid d)\right)}{\Phi^{-1}\left(\hat{F}_{D_1}(d)\right) - \Phi^{-1}\left(\hat{F}_{D_0}(d)\right)}$$

and

$$\hat{F}_{Y_1(d)}(y) = \Phi\left(\frac{\hat{a}_{d,y}}{\sqrt{1 + \hat{b}_{d,y}^2}}\right) \quad \hat{\rho}_d(y) = \frac{-\hat{b}_{d,y}}{\sqrt{1 + \hat{b}_{d,y}^2}}$$

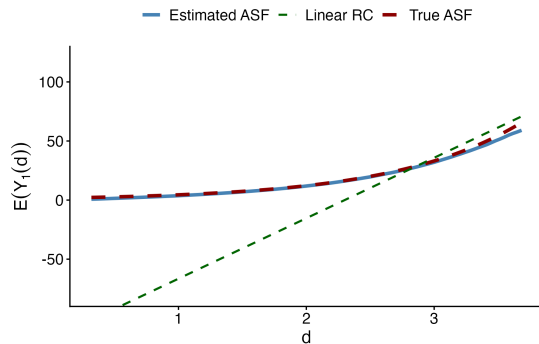
Illustration

DGP:

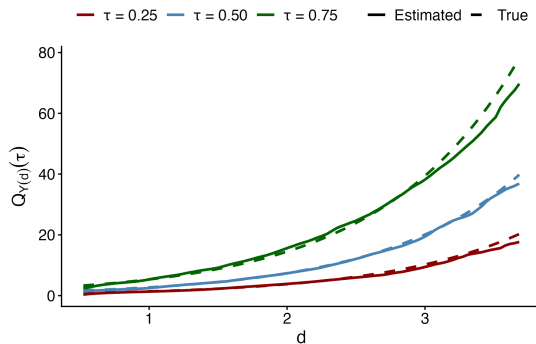
$$\begin{aligned}Y_{i0}(d) &= d + \varepsilon_{i0}, & D_{i0} &= 1 + \eta_{i0} \\Y_{i1}(d) &= \exp(d + \varepsilon_{i1}), & D_{i1} &= 2 + 2\eta_{i1}\end{aligned}$$

where $(\varepsilon_{it}, \eta_{it})$ follows a standard normal distribution with correlation $\rho = 0.5$

Illustration



Average structural function



Quantile structural function

► Correlation

Conclusion

This paper

- ▶ Study the nonlinear DiD models with continuous treatment in a repeated cross-section setting
- ▶ Identify the counterfactual outcome distribution by restricting local dependence of unobservables

Future work

- ▶ Extend to panel data
- ▶ Estimation and Inference
- ▶ Empirical Application

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